Food Remedy API

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Assigned Task.

This research will create a practical guide for deploying Food Remedy through Google Play Closed Testing, with access restricted to approved Food Remedy team members.

Tasks:

1. Research the current Google Play closed-testing process.
2. Document the required build, store, privacy and testing prerequisites.
3. Explain how to create the closed-testing track and allowlist testers.
4. Explain how approved users receive the opt-in link and install the app.
5. Include basic release, access-removal and rollback guidance.

Main objective

To develop a practical and easy-to-follow guide for securely deploying the Food Remedy Android application through Google Play Closed Testing, with access limited to approved project members.

Build prerequisites

Document the need for:

- A release-ready Android App Bundle, normally an .aab file
- A unique application package name
- A release version name and version code
- A signed release build
- Play App Signing
- A compatible Android target API level
- Removal of debugging settings and development credentials
- Testing the build before upload

1. Review of the Food Remedy Handover and Current Android Deployment Status

1.1. Overview

The Food Remedy project handover was reviewed to understand the application's current development stage, technical architecture, completed deployment activities, and remaining work required before Google Play Closed Testing.

Food Remedy is a mobile nutrition and food-assistance application designed to help users make healthier and safer food choices. Its main functions include barcode scanning, nutrition and allergen analysis, dietary suitability checks, personalised recommendations, shopping-list management, cart and checkout workflows, and user-profile management.

The application was developed using **React Native, Expo and TypeScript**, with **Firebase Authentication**, **Cloud Firestore**, **Firebase Storage** and **SQLite** supporting authentication, cloud storage and offline functionality. The project also includes a structured product-data pipeline for scraping, cleaning, enriching, validating and uploading food-product information.

1.2. Current project status

The handover describes Food Remedy as a strong minimum viable product that is approaching production readiness. The core user journey is largely operational, including:

- User registration, login and password recovery
- User profiles, demographic information and dietary preferences
- Product barcode scanning and searching
- Nutrition, ingredient and allergen information
- Personalised product insights
- Shopping-list, cart and checkout functions
- Firebase and SQLite data synchronisation
- Offline data caching and reconnection support

The previous team also completed frontend, backend, database and Firebase integration testing. **However, the handover makes clear that the project still requires additional production hardening and broader deployment testing before it can be considered fully production ready.**

1.3. Current Android deployment status

The handover confirms that the team:

- Used Expo for mobile development and application builds.
- Prepared Android application builds.
- Used Android Studio for emulator and APK testing.
- Conducted deployment-readiness checks.
- Tested Firebase infrastructure, authentication and Firestore integration.

However, the handover does not provide evidence that the application has been uploaded to Google Play Console or distributed through a Google Play testing track. The project resources section does not identify an active deployed frontend or backend, and no Google Play application, closed-testing track, tester list or opt-in link is documented. **Therefore, the current Android status is best described as build-tested and deployment-prepared but not yet confirmed as deployed through Google Play Closed Testing.**

To verify the current implementation, the GitHub repository was reviewed. The repository contains the Android application source code, Expo configuration (app.json), EAS build configuration (eas.json), Firebase configuration files, Firestore security rules, and project documentation. These files confirm that the application has the necessary technical foundation for generating Android builds and supporting deployment activities.

However, the review also found that several deployment activities are not yet confirmed within the repository. These include Google Play Console configuration, application signing, closed-testing setup, privacy declarations, and tester management. These activities are performed through external platforms such as Google Play Console and Expo EAS rather than being stored in the source code.

Therefore, the project is assessed as **partially ready for Google Play Closed Testing**. The application has an appropriate technical foundation, but deployment-specific configuration and verification are still required before the application can be distributed to testers.

2. Technical Guide to Set Up Google Play Closed Testing

The following steps provide a practical process for preparing and deploying the **Food Remedy Android application** through Google Play Closed Testing. The process begins with checking the existing application and finishes with managing testers and future test releases.

Step 1 – Prepare and run the Food Remedy application

Before preparing a Google Play release, the current version of Food Remedy should be successfully run and tested in the development environment. This provides a basic check that the application and its dependencies are working correctly.

```
cd Food-Remedy_API\mobile-app

Install dependencies:

npm install

Start the application:

npx expo start
```

The application should then be tested using an Android emulator or physical Android device.

Step 2 – Check the Android application configuration

The Android application information should be reviewed before creating the release. This is particularly important because information such as the package name identifies the application in Google Play.

Review:

mobile-app/app.json

A few items should be confirmed before Google Play Closed Testing:

1. The Android package name com.bryce_pillwein.foodremedy should be confirmed as the final package name before the first Google Play release.
2. The EAS project ownership should be verified to ensure that the correct Food Remedy or company account controls the build project.
3. The final release build should be tested to confirm that SQLite, Firebase, barcode scanning, and other core functions work correctly outside Expo Go.

Step 3 – Check the EAS production build configuration

Food Remedy uses **Expo Application Services (EAS)** to manage application builds. The production configuration should be checked to ensure that it is suitable for creating the Android release.

No major change to eas.json is required at this stage. However, the configuration file alone does **not confirm that the overall EAS setup is complete**. Before creating the Google Play build, the team should verify:

1. **The correct Expo account owns the EAS project.**

   From the mobile-app folder, run: **eas whoami**

   This shows which Expo account is currently logged in.

   Then run: **eas project:info.** Check that the project information matches Food Remedy and that the account belongs to the authorised project/company team.

2. **Android signing credentials are configured.**

   Run: **eas credentials --platform android.** Select the production profile when prompted.

   Review whether an Android keystore/upload key already exists. EAS can manage the Android signing credentials automatically, or the team can provide its own credentials. These credentials should be kept private and must not be stored in GitHub.

3. A production .aab can be generated successfully.

   Run: **eas build --platform android --profile production**

   The existing Food Remedy production profile should generate an Android App Bundle (.aab) because no APK build type has been specified.

   After the build finishes:

   - Check that the build status is Finished/Successful.
   - Download the .aab from the EAS build page.
   - Record the build version and date.

4. **The production build works correctly outside Expo Go.**

A production .aab cannot normally be installed directly on an emulator. For pre-release testing, first generate an installable APK using the existing preview profile:

eas build --platform android --profile preview

EAS should produce an installable Android build. Expo recommends APK builds for direct installation on Android devices and emulators

Step 4 – Create or configure Food Remedy in Google Play Console

The Food Remedy application must be registered in Google Play Console before a testing release can be distributed. This creates the Google Play application record that will be used for testing and future releases.

Step 5 – Create the Closed Testing track

The Closed Testing track provides the controlled environment where Food Remedy can be distributed without making it publicly available.

In Google Play Console, navigate to:

Test and release → Testing → Closed testing

Then:

1. Create a closed-testing track.
2. Give the track a clear name, such as **Food Remedy Team Testing**.
3. Configure the testing settings.

This track can then be reused for future Food Remedy test releases.

Step 6 – Create the approved tester group

Access to the application should be limited to approved Food Remedy team members. Google Play supports controlled tester groups using email lists or Google Groups.

The team should:

1. Create a tester list or select a Google Group.
2. Add approved testers.
3. Check that the correct Google account email addresses have been provided.
4. Save the tester configuration.

Step 7 – Upload and create the test release

The .aab generated through EAS is uploaded to the Closed Testing track to create the first test release.

The release becomes available to eligible testers after the required Google Play processing or review is completed.

Step 8 – Provide the opt-in link to testers

Approved testers need to join the testing program before they can install Food Remedy from Google Play. Google Play provides an opt-in link for this purpose.

Testers should:

1. Receive the opt-in link from the project team.
2. Open it using their approved Google account.
3. Join the testing program.
4. Follow the Google Play installation link.
5. Install Food Remedy.
6. Begin testing the application.

The opt-in link should only be distributed to the intended project members.

Step 9 – Collect testing results

Once the application is installed, testers should evaluate the main Food Remedy workflows and report any problems they identify.

Feedback should include basic information such as:

- Device used
- Android version
- Food Remedy version
- Feature being tested
- Problem identified
- Steps that caused the problem

This information can help the development team reproduce and resolve issues.

Step 10 – Maintain a release record

Each test release should be documented so future project teams can understand what was released, when it was tested, and what issues were identified.

A simple record can contain:

Information	**Example**
Application version	1.0.0
Build date	Date of build
Testing track	Food Remedy Team Testing
Tester group	Approved project members
Changes	Features and fixes included
Known issues	Outstanding problems
Status	Active / replaced